

Ruitang Yang, PhD

Quantitative Research Scientist | Scientific Computing, Geospatial Analytics and Probabilistic Modelling

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PROFESSIONAL SUMMARY

Quantitative research scientist specializing in glacier remote sensing and coupled numerical modelling. Developed reproducible workflows that transformed multi-source optical, SAR and stereo imagery into glacier-change observations and open datasets. Led the design and release of an open-source, observation-constrained Python framework for Linux/HPC that coupled climatic mass balance, ice dynamics and frontal ablation, demonstrated at regional scale and designed for global-scale applications.

TECHNICAL EXPERTISE

- **Scientific computing:** Python, Shell, Git, Linux, HPC; additional experience with MATLAB and C
- **Modelling and inference:** coupled process-based modelling, Bayesian calibration, data assimilation, ensemble simulation, scenario analysis, evaluation and uncertainty quantification
- **Geospatial and Earth observation:** QGIS, ArcGIS, ENVI and GAMMA; optical and SAR imagery, stereo-derived DEMs, laser altimetry and multi-source geospatial integration
- **Scientific software engineering:** modular workflow design, version control, documentation, validation, reproducibility, open-source release and research-data publication
- **Languages:** Mandarin Chinese (native); English (fluent)

PROFESSIONAL EXPERIENCE

- **Researcher, University of Oslo** Oslo, Norway | 2022–Present
 - Led development of a scalable, observation-constrained glacier-modelling and data-assimilation framework designed for regional-to-global applications and demonstrated across 71 marine-terminating glaciers in Svalbard.
 - Built reusable Python and Shell pipelines on Linux/HPC for multi-source geospatial data integration, Bayesian calibration, ensemble simulation, model evaluation, uncertainty quantification, and projections through 2100.
 - Delivered documented public software, four first-author studies, and four public datasets with persistent identifiers and traceable provenance.
 - Communicated modelling methods, assumptions, uncertainty, and results to international collaborators through technical documentation, publications, and presentations.
- **Senior High School Mathematics Teacher and Class Adviser, Longxi No. 1 Middle School** Gansu, China | 2011–2016
 - Explained complex quantitative concepts to diverse learners and designed structured approaches to assessment, problem solving, and progress evaluation.
 - Mentored and coordinated a class of approximately 60 students for three years, working with families, colleagues, and school leadership.

SELECTED TECHNICAL PROJECTS

- **Integrated Glacier Modelling with Joint Bayesian Calibration**

Integrated OGGM, PyGEM, and SERMeQ to couple climatic mass balance, glacier geometry evolution, ice dynamics, and frontal ablation within a modular framework designed for regional-to-global applications and calibrated jointly against multiple observational constraints.
- **Multi-Source Geospatial Analysis**

Integrated optical imagery for glacier mapping, SAR-derived surface velocities, stereo-derived DEMs, and modelled runoff to quantify multi-decadal glacier change and seasonal glacier-runoff relationships in the Kenai Peninsula, Alaska.

EDUCATION

- **PhD in Physical Geography, University of Chinese Academy of Sciences** China | **2022**
Visiting PhD researcher at the University of Alaska Fairbanks, USA (18 months), and the University of Oslo, Norway (12 months)
- **MSc in Mathematics Education, Shaanxi Normal University** China | **2015**
- **BSc in Mathematics and Applied Mathematics, Shaanxi Normal University** China | **2011**

SELECTED DELIVERABLES

- **Research software**
Led the design and development of a documented framework for observation-constrained simulation and projection of marine-terminating glacier change, coupling climatic mass balance, ice dynamics and frontal ablation. | [Software DOI](#)
- **First-author publications**
Authored three peer-reviewed first-author publications and one first-author preprint spanning glacier remote sensing, glacier dynamics, glacier-runoff relationships and coupled glacier modelling. | [Google Scholar](#)
- **Open datasets**
Published four reusable datasets (three regional observational datasets and one model-projection dataset) with persistent identifiers and traceable provenance. | [Dataset records](#)

LEADERSHIP AND COMMUNICATION

- **GlacierMIP member and subgroup Co-Chair** **2022–Present**
Co-lead the development of standardized experimental protocols for GlacierMIP4 and contribute to international model coordination.
- **Workshop and Seminar Organization** **2025**
Organizing committee member for the CryoDA workshop (Data Assimilation for Cryospheric Applications) and the Glacier Modelling workshop at the University of Oslo.
- **Seminar Organization** **2022**
Organized a weekly research seminar for one semester in the Department of Geosciences, University of Oslo.
- **Scientific Peer reviewer** **2024–Present**
Reviewer for Journal of Glaciology, Advances in Climate Change Research, and GIScience & Remote Sensing; evaluator for the AGU Outstanding Student Presentation Award program.
- **Research Mentoring** **2017–2018**
Co-mentored an MSc thesis in glacier remote sensing.

RECOGNITION

- **Top Cited Article 2022–2023** **2024**
Journal of Geophysical Research: Earth Surface.
- **Best Student Oral Presentation** **2021**
International Glaciological Society Nordic Branch Symposium 2021

SELECTED FIELD EXPERIENCE

- **Field Assistant** | Kennicott Glacier, Alaska, USA **2018 & 2020**
Ablation-stake drilling, automatic weather-station and time-lapse camera installation, and snow/water sampling; four weeks across two campaigns.
- **Field Assistant** | Halong Glacier, Qinghai-Tibet Plateau, China **2019**
Ablation-stake drilling, automatic weather-station installation; three weeks.
- **Snow and UAV Field Campaign** | Zhangjiakou, China **2017**
Snow-pit measurements, melt-stake drilling, time-lapse monitoring, and UAV surveys; 66 days.